Shahnaaz Parveen

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.50
SRI CHAITANYA JUNIOR COLLEGE	Class XII (Board of Intermediate Education-BIEAP)	2020	96.20%
S.F.S HIGH SCHOOL	Class X (BOARD OF SECUNDARY EDUCATION-AP)	2018	10.00%

SCHOLASTIC ACHIEVEMENTS

- **NPTEL Elite Certified** in *VLSI Design Flow: RTL to GDS*
- Awarded a **Certificate of Merit by ISRO** for successfully completing the *Overview of Space Science and Technology* training programme with an **A+ grade**.

POSITIONS OF RESPONSIBILITY

- **Class Representative**, M.Tech, Semester I | Aug 2025 - Dec 2025
- **Placement Coordinator**, M.Tech Placement Session | 2026-Present
- **Teaching Assistant**, Department of Electrical Engineering, IIT Hyderabad -- assisted in *EE1080: Probability and Random Processes* and *EE3900: Linear Systems and Signal Processing*
- **Head Coordinator**, Technical Fest, Department of Electronics and Communication Engineering, Andhra University

TECHNICAL SKILLS

- **EDA & Synthesis Tools:** Intel Quartus Prime, Xilinx Vivado
- **HDL & Programming:** Verilog, C, C++, MATLAB, Data Structures and Algorithms
- **Simulation & Verification:** ModelSim, QuestaSim
- **Protocols**: UART, SPI, I²C, AHB, AXI

RELEVANT COURSES

- Random Variable and Stochastic Processes
- Principles of Digital Communication
- Wireless Communications
- Information theory
- Digital Electronics
- Advanced Digital Signal Processing
- Digital IC design
- Analog Electronics Circuits
- Pattern Recognition and Machine Learning
- Matrix Theory

PROJECTS

M.Tech Thesis --Hardware Implementation of OTFS Synchronization *Ongoing*

IIT Hyderabad | 2026-Present

- Developed a **MATLAB-based comb-preamble algorithm for Symbol Timing Offset (STO) estimation in OTFS systems**.
- Implemented and evaluated Joint Time Synchronization and Channel Estimation for OTFS for **delay and Doppler**

estimation as part of the OTFS synchronization framework.
- Currently working toward **FPGA implementation of the proposed synchronization algorithm**, targeting hardware-efficient and real-time processing.
- Tools: MATLAB, Intel Quartus Prime, Model Sim
- Hardware: Xilinx Zynq UltraScale+

5-Stage Pipelined RISC-V Processor

- Designed and implemented a **5-stage pipelined RISC-V (RV32I) processor** with IF, ID, EX, MEM, and WB stages, including dedicated pipeline registers and control logic.
- Implemented **data-hazard handling** using EX/MEM and MEM/WB forwarding, load-use hazard detection, pipeline stalling, and branch flushing for control hazards.
- Performed **Static Timing Analysis (STA)** and **resource utilization analysis ,** targeting Cyclone-V in Intel Quartus Prime to evaluate critical paths, timing slack, Fmax, logic utilization, and register usage.
- **Tools:** Verilog, Intel Quartus Prime, ModelSim/QuestaSim

2-Way Set-Associative Write-Back Cache

- Designed a **2-way set-associative cache** with 4-word lines, supporting **write-back, write-allocate, dirty-bit management, and LRU replacement**.
- Developed an **FSM-based cache controller** handling hits/misses, line refill, dirty-line eviction, and memory transactions.
- Built a **self-checking testbench** for functional verification and performed **STA and resource utilization analysis in Quartus** targeting Cyclone-V to analyze critical paths, setup/hold timing, and slack.
- **Tools:** Verilog, Intel Quartus Prime, ModelSim/QuestaSim

Pipelined CORDIC Accelerator

- Designed a **16-stage pipelined CORDIC accelerator** supporting rotation and vectoring modes for trigonometric and Cartesian transformations using fixed-point arithmetic.
- Developed a **parameterized RTL datapath** with guard-bit extension, quadrant correction, gain compensation, saturation, and configurable precision.
- Verified RTL outputs against **MATLAB reference models** and performed **STA and resource analysis** on Intel Cyclone V, evaluating critical paths, timing slack, Fmax, and FPGA utilization.
- **Tools:** Verilog, MATLAB, Intel Quartus Prime, ModelSim

32-Point Radix-2 DIT FFT Processor

- Designed and implemented a **32-point radix-2 Decimation-in-Time (DIT) FFT processor** using modular butterfly units, pipelined datapaths, and dedicated **twiddle-factor ROMs** for complex-valued frequency-domain computation.
- Developed **fixed-point Q8.8 arithmetic** with multi-stage pipelining, memory modules, and control logic to sequence butterfly operations and intermediate data.
- Developed a **MATLAB-based verification framework** to compare RTL FFT outputs against MATLAB's floating-point FFT, evaluate per-bin error and numerical accuracy, and performed **Static Timing Analysis (STA)** and Resource utilization analysis targeting Cyclone-V to analyze critical paths, setup/hold timing, slack, and maximum operating frequency.
- **Tools:** MATLAB, Intel Quartus Prime, ModelSim/QuestaSim

EXTRACURRICULAR

Designed e-posters and promotional creatives for the **National Conference on Communications (NCC) 2026**, IIT Hyderabad, supporting conference branding and digital outreach.

Contributed to event branding and visual communication by designing promotional creatives, including the main and event posters, for the B.Tech Technical Fest at Andhra University.

CERTIFICATIONS

- NPTEL Elite Certification- RTL to GDS
- Earned a Certificate of Merit from ISRO -START programme
- Udemy certification : ASIC Synthesis-STA-Physical Design (PD)

INTERNSHIPS

Embedded Systems & IoT Intern - CEMS*(Sept-Nov 2022)*

- Worked with Arduino UNO, ESP32, and IoT/cloud platforms including Blynk and Tinkercad.

BHEL - CNC Systems Intern *(May-June 2023)*

- Gained industry exposure to CNC and PLC-based automation systems and their applications across industrial sectors.

Rudhaa SkillForge Technologies - VLSI Design Engineer Intern (AICTE) *(Feb-Mar 2024)*

- Worked with RTL design using Verilog HDL, developing and simulating digital hardware modules.

VLSI Mentorship Program -- Launched Global

- Completed a structured mentorship program focused on VLSI design and RTL concepts.